

28/08/26

Signal Processing

Recap: LTI systems, impulse response, convolution sum, convolution in time $\Rightarrow$ multiplication of DTFT

Examples:

Our favourites: $h_{\text{diff}}[n] = \delta[n] - \delta[n-1]$ ✓

$$h_{\text{avg}}[n] = \frac{1}{2} (\delta[n] + \delta[n-1])$$

Signals: $x[n] = u[n]$ ✓

$$x[n] = \cos(\pi n)$$

$$\Rightarrow y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k] = \sum_{k=-\infty}^{\infty} x[k] \cdot T\{\delta[n-k]\} \leftarrow \}$$

$\Rightarrow$ LTI system, $h[n] = T\{\delta[n]\}$ completely describes the system.

$$\Rightarrow y[n] = x[n] * h[n]$$

$\hookrightarrow$ convolution symbol

$$\Rightarrow \text{LTI systems: } y[n] = x[n] * h[n] \Rightarrow Y(e^{j\omega}) = X(e^{j\omega}) \cdot H(e^{j\omega})$$

$$\Rightarrow Y(z) = X(z) \cdot H(z)$$

(careful about ROC)

$$\Rightarrow Y[k] = X[k] \cdot H[k]$$

(need to be careful about periodicity).

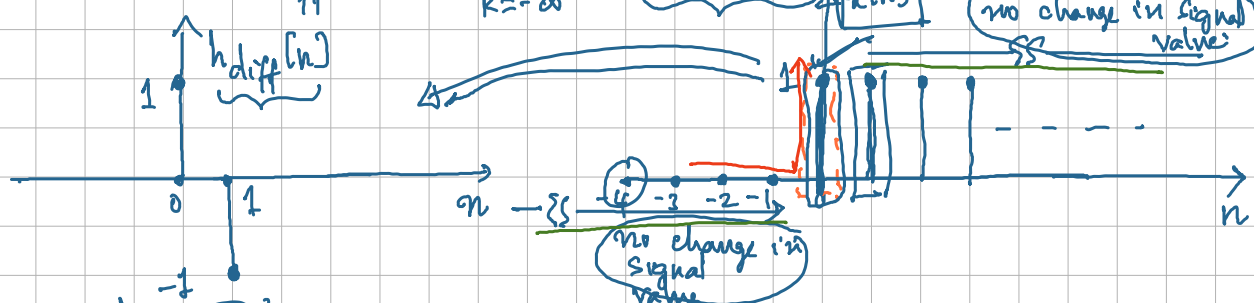
particularly useful property in practise.

Q. Find the response of $h_{\text{diff}}[n]$ to the unit step signal. $x[n] = u[n]$

A. Note that $h_{\text{diff}}[n]$ is an LTI system. Please verify this claim.

$$y[n] = x[n] * h_{\text{diff}}[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot T\{\delta[n-k]\}$$

Ex 1.



$$|H_{\text{diff}}(e^{j\omega})| = \left| \sin\left(\frac{\omega}{2}\right) \right|$$

This is a high-pass filter since it attenuates low frequencies and allows high freqs.

$$n = -4 ; \quad \underbrace{x[k]}_{x[-4]} \cdot \underbrace{\delta[-4-k]}_{\delta[-4-(-4)]} = \underbrace{x[-4]}_{0} \cdot \underbrace{\delta[0]}_{1} = 0$$

$$n = -3 ; \quad \underbrace{x[k]}_{x[-3]} \cdot \underbrace{\delta[-3-k]}_{\delta[-3-(-3)]} = 0$$

$$\begin{aligned} n = 0 ; \quad & x[k] \cdot \sum \delta[n-k] \\ &= x[0] \cdot \sum \delta[n] \quad \leftarrow \text{impulse} \\ &= x[0] \cdot \underbrace{\sum \delta[n]}_{h[n]} \end{aligned}$$

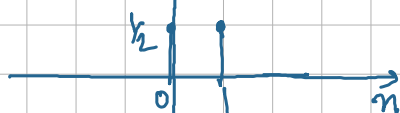
Let us compute $y[0] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[0-k]$

$$\begin{aligned} &= 1 \\ y[3] &= \sum_{k=-\infty}^{\infty} x[k] \cdot h[3-k] \\ &= 0 \end{aligned}$$

$$\therefore y[n] = \begin{cases} 0, & n \neq 0 \\ 1, & n = 0 \end{cases}$$

Ex 2:

$$h_{\text{ave}}[n] = \frac{1}{2} (\delta[n] + \delta[n-1])$$



$$y[-1] = 0$$

$$y[1] = 0$$

$$y[2] = 0$$

$$y[n] = 0 \quad \text{for all } n$$

